

Theoretical Assignments

1. Compare SQL and NoSQL Databases

SQL Databases :

SQL stands for **Structured Query Language**.

These databases store data in **tables (rows & columns)**.

Examples:

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

Characteristics:

- Fixed schema (structure defined before inserting data)
- Uses SQL language
- Supports ACID properties
- Best for structured data

NoSQL Databases :

NoSQL means **Not Only SQL**.

These databases store data in flexible formats like:

- Document
- Key-Value
- Graph

- Column-based

Examples:

- MongoDB
- Cassandra
- Redis
- Neo4j

Characteristics:

- Flexible schema
- Handles large scale data
- Better horizontal scaling
- Suitable for unstructured / semi-structured data

2. Presentation: Normalization vs. Denormalization

What is Normalization?

Normalization is the process of organizing data in a database to:

- Reduce redundancy
- Avoid data anomalies
- Improve data integrity

It divides large tables into smaller related tables.

Types of Normal Forms:

1. 1NF (First Normal Form) – No repeating groups
2. 2NF (Second Normal Form) – No partial dependency
3. 3NF (Third Normal Form) – No transitive dependency

Advantages of Normalization:

- Eliminates duplicate data
- Saves storage space
- Maintains data consistency

Disadvantages of Normalization:

- More tables
- Complex queries (more joins)
- Slightly slower read performance

What is Denormalization?

Denormalization is the process of combining tables to:

- Reduce joins
- Improve read performance

It intentionally adds redundancy.

Advantages of Denormalization:

- Faster read queries
- Simple queries
- Better performance in reporting systems

Disadvantages of Denormalization:

- Data redundancy
- Higher storage usage
- Risk of data inconsistency